Problem Definition and Research Focus

Explaining employee attrition models, and testing whether the explanations hold

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1. Background

The cost of replacing an experienced employee is expensive: recruitment costs, time of hiring manager, productivity loss during the time the position is unfilled, and the loss of organisation knowledge. The cost of a retention conversation is a small fraction of replacing a person and predicting who is likely to leave has become one of the most studied issues in HR analytics.

A literature review of the 2019–2024 years reveals that reasonably good predictions of attrition can be obtained most frequently using tree ensembles. The open problem isn't prediction. What's next is what happens.

2. Problem definition

A list of at-risk employees is not actionable all by itself. That list needs to be held by an HR manager, who in turn must decide who to approach, what to offer to do and why, in the event that it is questioned. Two employees can have the same risk score for two different reasons: one sustained overtime – this can be changed and the other because of their age and marital status – this is a factor they can't, and shouldn't, change. It is only because of a reason that they differ. Explainability is thus a functional requirement, not something optional.

The usual approach is to fit a black box model, add a post hoc explainer (which is almost invariably SHAP or LIME) and publish the feature importances obtained as the study's result. This project is designed around 2 challenges to not stopping there:

1. **The explanations are rarely tested.** A post-hoc explanation approximates the model, not the world. LIME samples a random neighbourhood, so two runs on the same prediction can disagree. SHAP is deterministic for a fixed model, but the model is itself a sample artefact, so a slightly different set of employees produces a different ranking. Applied papers publish a single bar chart from a single fit and seldom ask whether it would survive being computed again.

2. **The interpretable alternative is rarely taken seriously.** In the case of decisions that matter, Rudin (2019) maintains that such decisions should be made with the help of an intrinsically interpretable model where possible, as it is an explanation and is incapable of contradicting itself. The argument is most likely to hold true in the regime of retention decisions, which involve pay, progression and treatment at work, as well as the IBM HR dataset (1,470 rows, 34 features, structured tabular data). But studies of attrition usually only incorporate the interpretable model as a base case to be outdone.

Problem statement. HR attrition studies routinely publish post-hoc explanations whose stability is untested, and rarely check whether a model that needs no explanation would perform just as well. Without that evidence, neither the feature ranking nor the individual risk explanation can be defended when an HR decision based on it is challenged.

3. Research focus

The project treats the **explanation itself as the object under test**, rather than the model's accuracy. Three research questions guide the work:

	Research question	Focus
RQ1	Which employee characteristics are the strongest predictors of attrition, and how consistent are they across models and across explanation methods?	Consistency of the feature ranking
RQ2	How does an intrinsically interpretable model compare with a black-box model on predictive performance, and what is the cost of interpretability?	Performance cost of a glass-box model
RQ3	To what extent can these explanation outputs be translated into actionable, defensible guidance for an HR manager?	Stability and defensibility of explanations

RQ1 and RQ3 are deliberately framed around *consistency* and *defensibility* rather than importance alone: the question is not only which features a model nominates, but whether the nomination survives being asked again.

3.1 What the study does

- Trains **five classifiers** — three glass-box (logistic regression, pruned decision tree, explainable boosting machine) and two black-box (random forest, XGBoost) — on **one shared 51-column feature encoding**, so disagreement between their explanations is genuine and not an encoding artefact.
- Compares **four class-imbalance strategies** inside cross-validation, and evaluates with PR-AUC, recall and F1 at a chosen threshold rather than accuracy (a model that predicts "nobody leaves" already scores 83.9% accuracy).

- Explains every model with **SHAP and LIME**, and compares the two employee by employee on the same model.
- Runs **three stability stress tests** on the explanations: refitting on bootstrap resamples, perturbing the continuous inputs by 0.1 SD, and re-running LIME with nothing changed but the random seed.
- Performs a **subgroup audit** by gender and age band.
- Produces a **reproducible pipeline** (single command, single seed, checksummed data) and a **practitioner guidance note**.

3.2 Scope and boundaries

In scope: binary attrition classification on the IBM HR Analytics dataset; the five models above; SHAP and LIME explanations; agreement and stability analysis; a subgroup audit.

Out of scope: live deployment; collection of new employee data; causal estimation of interventions (nothing here shows what *would* happen if overtime were reduced); time-to-event modelling of *when* someone leaves; and recommendations about identified individuals. The dataset is IBM sample data with no real employees, which keeps the ethical position straightforward but rules out claims about any real workforce.

4. Expected contribution

A like-for-like comparison of the glass-box and black-box attrition models, a quantitative stability analysis of the explanations provided by the models, a direct comparison between SHAP and LIME on the individual employee level, evidence on the cost of interpretability on this dataset, and guidance for HR practitioners based on the stability results as opposed to a single ranking of the features.